Apriori Algorithm — ~~Mean~~ Finding frequent itemsets by confined candidate generation

Example: All electronics transaction database 'D'

TId	Item Id's
T1	I1, I2, I5
T2	I2, I4
T3	I2, I3
T4	I1, I2, I4
T5	I1, I3
T6	I2, I3
T7	I1, I3
T8	I1, I2, I3, I5
T9	I1, I2, I3

Step 1:- In the first iteration of the algorithm each item is a member of the set of candidates one item set C1.

Step-2:- Suppose that min support count required is 2 i.e min support = 2. The set of frequent one item set L1 can be then determined. It consists of 1 item set satisfying min support i.e scan-D for count of each candidate

Items	Support Count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

C1

Step-3:- To discover the set of frequent two item sets L2 the algorithm uses the JOIN $L_1 \bowtie L_1$ to generate a candidate set of 2 itemsets C2. i.e compare cand. support count with minimum support count.

Items	Support Count
I1	6
I2	7
I3	6
I4	2
I5	2

L1

→

C2

Items	Support Count
{I1, I2}	4
{I1, I3}	4
{I1, I4}	1 ⊗
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2
{I3, I4}	0 ⊗
{I3, I5}	1 ⊗
{I4, I5}	0 ⊗

Step-4:- Next the transactions in D are scanned and the support count of each candidate support count is accumulated.

* The set of frequent 2-Item sets L2 is then determined consisting of those candidate two item set in C2 having minimum support. i.e L2.

Items	Supp. count
{I1, I2}	4
{I1, I3}	4
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2

L2

Step-5 :- The generation of the set of candidates 3-Item set C3 from the join set C3 is L2 ⋈ L2.

Itemset	Support count
{I1, I2, I3}	2
{I1, I2, I5}	2
{I1, I2, I4}	1 ⊗
{I1, I3, I5}	1 ⊗
{I2, I3, I4}	0 ⊗
{I2, I3, I5}	1 ⊗
{I2, I4, I5}	0 ⊗

C3

Step-6

Compare cand supp count with min supp count

Itemset	Supp count
{I1, I2, I3}	2
{I1, I2, I5}	2

L3

Step-6 :- Generate from L3 to C4 (4-Items set ID's)

Item	Support count
{I1, I2, I3, I5}	1 ⊗

L4 = $\boxed{\phi}$

L4 doesnot contain any Itemset then apporiori algorithm is terminated

i.e $L_K = \phi$ L_4 does not support on any candidate item sets $L_{K-1} \Rightarrow L_3$ frequent Item sets

Note: When you were $L_4 = \phi$ the algorithm terminated the Itemsets in L_3 are frequent.

$\{I1, I2, I3\}, \{I1, I2, I5\}$

Step 7 $C_3 = L_2 \bowtie L_2$

(b) prune using the apporiori property - all non empty subsets of a frequent Itemset must also be frequent.

* The two item subsets $\{I1, I2, I3\}$ are $\{I1, I2\}, \{I1, I3\}, \{I2, I3\}$ all two item subsets of $\{I1, I2, I3\}$ are members of L_2. therefore keep $\{I1, I2, I3\}$ in C_3.
* The two Item subsets of $\{I1, I2, I5\}$ are $\{I1, I2\}, \{I2, I5\}, \{I1, I5\}$. All two Item subsets of $\{I1, I2, I5\}$ are members of L_2. therefore keep $\{I1, I2, I5\}$ in C_3.
* The two itemsubsets of $\{I1, I3, I5\}$ are $\{I1, I3\}, \{I1, I5\}, \{I3, I5\}$. $\{I3, I5\}$ is not a member of L_2 and so it is not frequent therefore remove $\{I1, I3, I5\}$ from C_3
* The two itemsubsets of $\{I2, I3, I4\}$ are $\{I2, I3\}, \{I2, I4\}, \{I3, I4\}$. $\{I3, I4\}$ is not a member of L_2 so it is not frequent therefore remove $\{I2, I3, I4\}$ from C_3.
* The two item subsets of $\{I2, I3, I5\}$ are $\{I2, I3\}, \{I3, I5\}, \{I2, I5\}$. $\{I2, I5\}$ is not a member of L_2 and so it is not frequent remove $\{I2, I3, I5\}$.

* The two item subsets of {I2, I4, I5} are {I2, I4}, {I2, I5}, {I4, I5}. {I4, I5} is not a member of L2 remove the {I2, I4, I5} from C3

c) Therefore C3 = {I1, I2, I3}, {I1, I2, I5} after pruning.

Step-7:- The transactions in 'D' are scanned to determine L3 consisting of those candidate 3-Item sets in C3. having min support

Step 8:- The algorithm uses L3 ⋈ L3 to generate candidate set of 4-Item sets C4.

{I1, I2, I3, I5} is pruned because its subset {I2, I3, I5} is not frequent

∴ C4 = ∅ and algorithm terminates having found all of the frequent itemsets.

Frequent Itemset Mining Methods

Three approaches

1. Apriori Algorithm
2. FP-growth Algorithm
3. Frequent pattern mining with vertical data format

2. FP-growth Algorithm :-

FP- stands for frequent pattern growth algorithm. the transactions < T1, T2, . . . ,T9 > &

frequent patterns are constructed tree pattern

Two steps of FP-growth algorithm

(i) FP-Tree construction

(ii) frequent pattern (or) Item sets

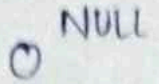
(i) FP-Tree Construction :-

Transaction ID	List of Items
T1	I2, I1, I5
T2	I2, I4
T3	I2, I3
T4	I2, I1, I4
T5	I1, I3
T6	I2, I3
T7	I1, I3
T8	I2, I1, I3, I5
T9	I2, I1, I3

Frequent Itemset

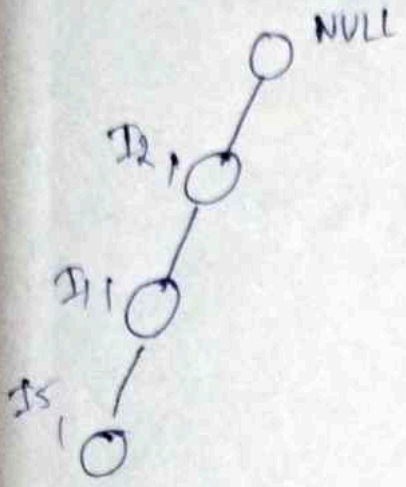
List of Items I1, I2, I3, I4, I5

a) Initially root Node

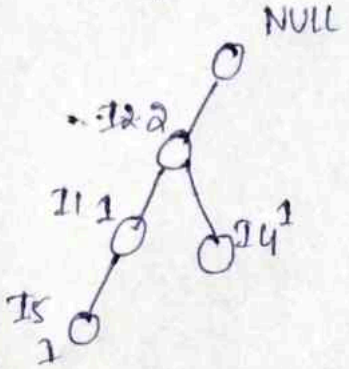


b) After Reading Transaction T1

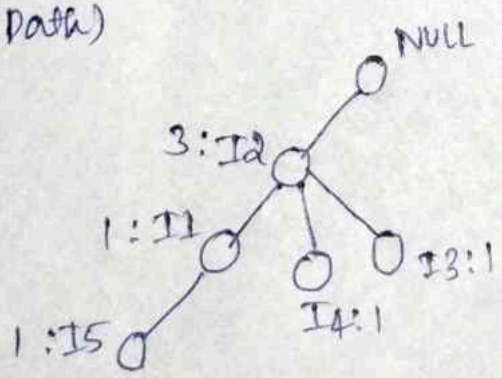
I2, I1, I5



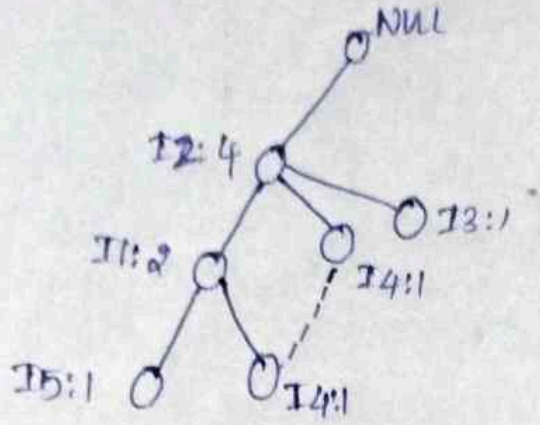
c) After reading T2 (we need to consider another path)



d) After reading Transaction T3 (we need to consider another path)



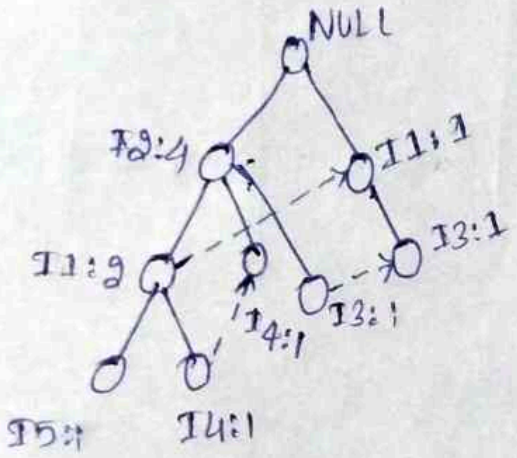
e) After reading T4 (we need to consider another Path)



If two nodes are same nodes that are connected with dotted lines

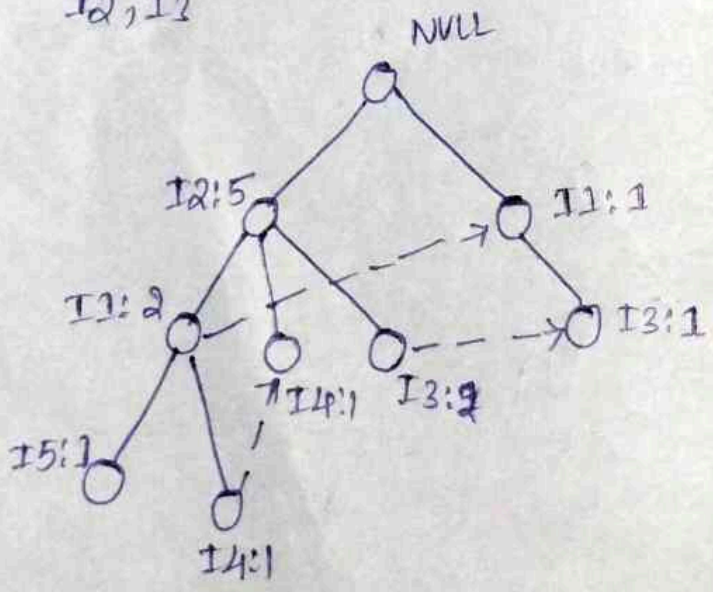
f) After reading T5 (we need to consider another path)

I1, I3



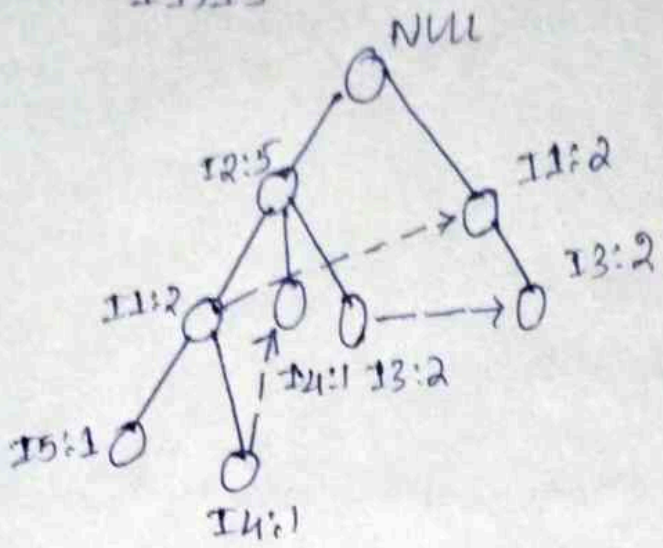
g) After reading T6

I2, I3



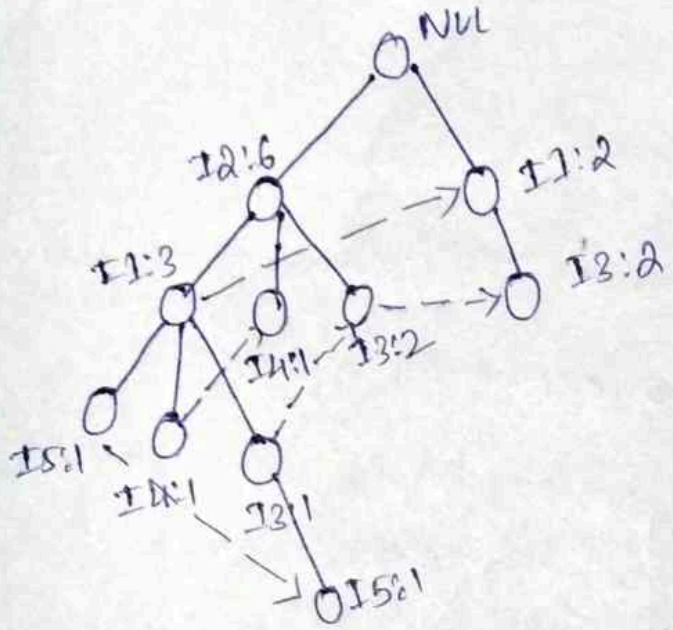
h) After Reading T7

I1, I3



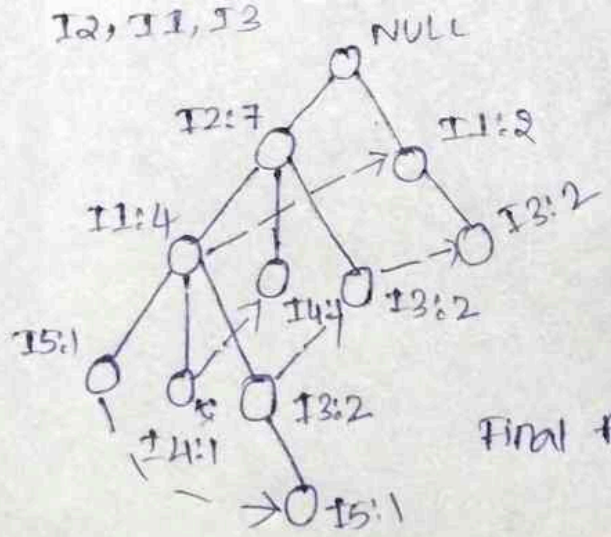
i) After Reading T8

I2, I1, I3, I5



j) After reading T9

I2, I1, I3



Final FP Tree construction

Items	Support Count
I1	6
I2	7
I3	6
I4	2
I5	2

Minimum support count is 2.

* We consider all the items. support count is arranged in descending order-

Items	Supp. Count
I2	7
I1	6
I3	6
I4	2
I5	2

Step-2:- finding frequent pattern / Itemsets:-

Finding frequent pattern which item has least count that item has consider I5

Item	condition pattern base	conditional FP Tree	Segment patterns Generated
I5	{I2:I1:1} {I2:I1:I3:1}	{I2:2, I1:2} {I2,I1} join with I5	{I2,I5}:2 I1,I5:2 I1:I2:I5:2}
I4	{I2;I1:1}{I2:1}	{I2:2} Joined with I4	{I2,I4:2}
I3	{I2,I1:2}{I2:2} {I1:2}	{I2:4}{I1:2} I1:2 min(4,2,2)=2	{I2,I3:4 I1,I3:4 I1,I2,I3:2}

* Conditional pattern base count value >=2 condition true then findout list of items in the count value otherwise the list of item are eliminated.

I2

Both are different we cannot combine

I1	{I2:4}	{I2:4}	{I1,I2:4}

(3) FP Mining with Vertical Data format :-

Tid	List of Items
T100	I2, I1, I5
T200	I2, I4
T300	I2, I3
T400	I2, I1, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I2, I1, I3, I5
T900	I2, I1, I3

* It is a ECLAT algorithm

ECLAT → Equivalent Class Transformation

* It is a BFS approach

1 - Item set

List of Items	TID	Cand Support cnt.
{I1}	{T100, T400, T500, T700, T800, T900}	6
{I2}	{T100, T200, T300, T400, T600, T800, T900}	7
{I3}	{T300, T500, T600, T700, T800, T900}	6
{I4}	{T200, T400}	2 *
{I5}	{T100, T800}	2 *

(1) If TID >= min support condit^n is true the Item sets are frequent

(2) If TID not greater than equal to 2 condit^n is false then ~~frequent~~ the Items sets are infrequent

infrequent Items are eliminated in the

2- Itemsets

Itemset	TID	
{I1, I2}	{T100, T400, T800, T900}	4
{I1, I3}	{T500, T700, T800, T900}	4
{I1, I4}	{T400}	1 (x)
{I1, I5}	{T100}	1 (x)) x
{I2, I3}	{T300, T600, T800, T900}	4
{I2, I4}	{T200, T400}	2
{I2, I5}	{T100, T800}	2
{I3, I4}	∅	∅ (x)
{I3, I5}	{T800}	1 (x)
{I4, I5}	∅	∅ (x)) x

3- Itemset

Itemset	TID	Supp Count
{I1, I2, I3}	{T800, T900}	2
{I1, I2, I4}	{T400}	1 (x)) x
{I1, I2, I5}	{T100, T800}	2
{I2, I3, I4}	∅	0 (x)
{I2, I3, I5}	{T800}	1 (x)
{I2, I4, I5}	∅	0 (x)) x

4- Itemset

{I1, I2, I3, I5}	{T800}	1 ⊗

4 - Itemset Vertical Data-format

* The Itemset is a Infrequent itemset then go for previous step

Itemset	TID
{I1, I2, I3}	{T800, T900}
{I1, I2, I5	{T100, T800}

* These two are frequent itemsets
* These algorithm uses intersection based approach.
* Intersection based approach means that intersection between the two items what are transactions are prese